

# Real-Time Social Media Mining and Trust-Aware Sentiment Analysis

Using Large Language Models for Retail Feedback Optimization

**BITS ZG628T · Dissertation**

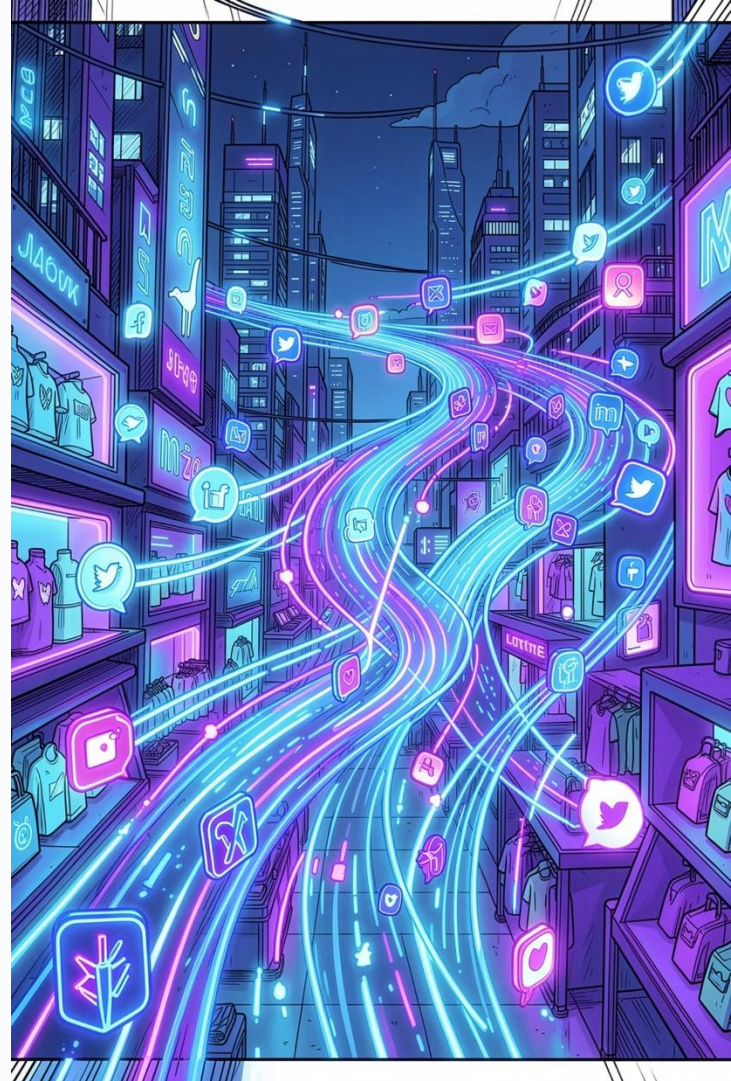
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**Program:** MTech · AI & Machine Learning

**Organization:** Walmart Global Tech · GIF



# What We're Building & Where We Are

An **automated, trust-aware pipeline** that tracks Walmart-related conversations on Reddit, uses **LLMs** to extract sentiment, aspects & emerging themes, and surfaces them in a **human-reviewed dashboard** so Ops · PR · Product teams can respond at social speed.

## 01 · Listen

25 Walmart / retail subreddits · hourly · Arctic Shift API.

## 03 · Review

Analysts validate, correct & prioritize in a Kanban board.

## 02 · Analyze

ModernBERT · DeBERTa-v3 · Gemma 3 vision · Trust score.

## 04 · Act

Route to Ops · PR · Product via P1 / P2 / P3 with SLAs, Post life cycle, inference for Walmart, review, replay etc .

### ✓ Part A · Mid-Semester

- ✓ End-to-end 6-stage pipeline (Ingest → Alert)
- ✓ Vision pipeline — Gemma 3 4B on image-only posts
- ✓ Trust score — Metadata · Dedup · LLM credibility
- ✓ ModernBERT fine-tune — Macro F1 0.76 (vs RoBERTa 0.63)
- ✓ Real-time dashboard — 22 API endpoints + WebSocket
- ✓ Notification system — P1/P2/P3 (Email · Slack · Teams) —SMTP finding out

### ● Part B · Post Mid-Semester

In progress

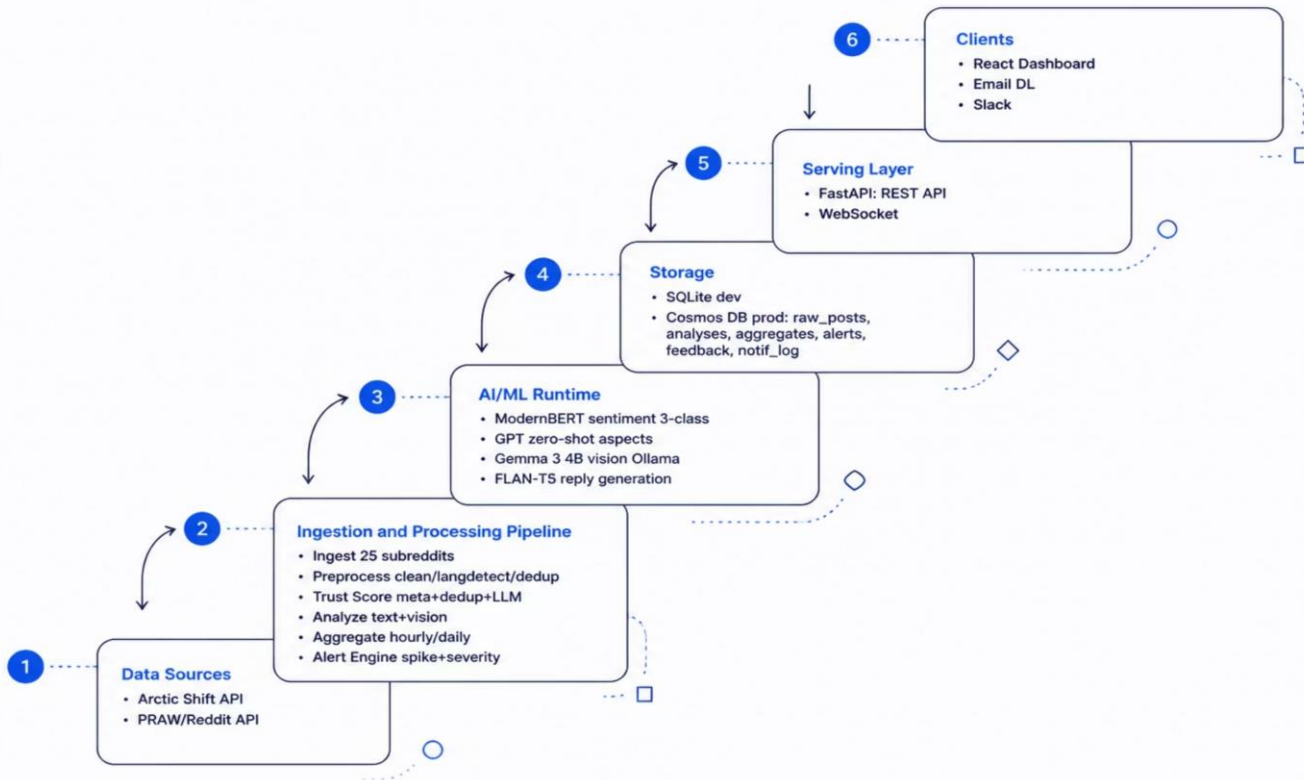
- ☐ Review & Validate — Human-in-the-Loop workflow
- ☐ Post Explorer — Multi-facet filtering & search
- ☐ Kanban Lifecycle — Triage → In-Review → Resolved
- ☐ Insights & Competitor — Cross-brand comparison
- ☐ Smart Reply Composer — Dual-draft (Rule + LLM)
- ☐ Learning loop — Feedback → future retraining

Overall Dissertation Progress Phase 1 of 2 - ~55% complete

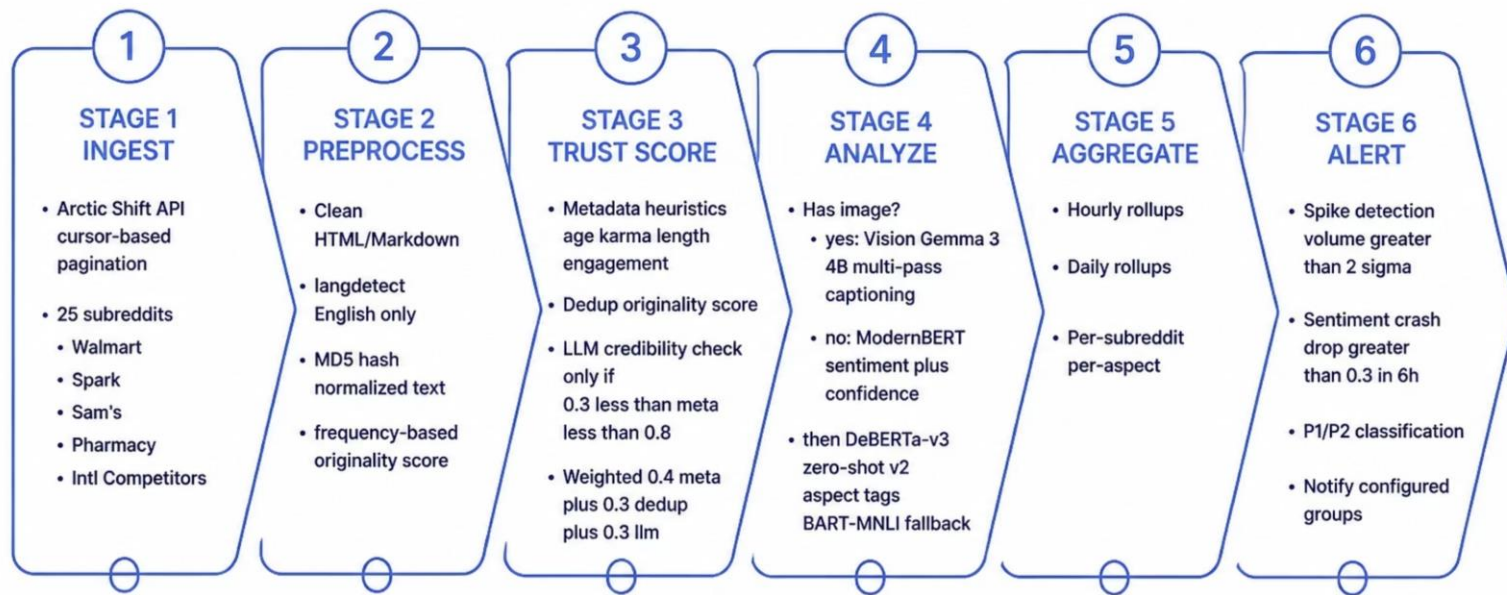


# High-Level System Architecture

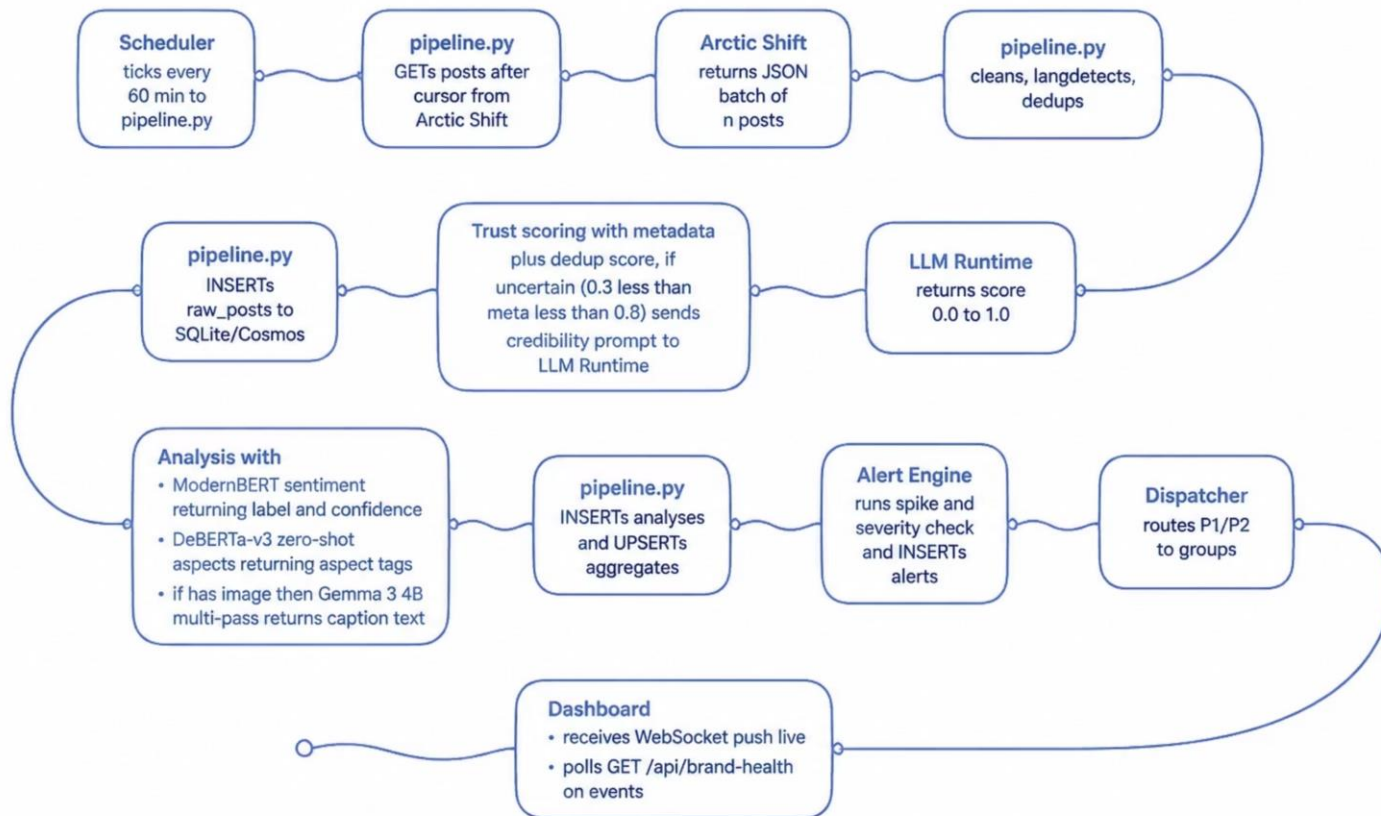
Layered view — external sources → pipeline core → AI runtime → storage → serving → clients



## 6-Stage Pipeline — Detailed Flow



## Request Sequence – End-to-End



# Data Ingestion — Where the Plan Met Reality

How we replaced PRAW with the Arctic Shift API and turned a broken plan into a stable 60-minute ingestion loop

## Initially Planned

### PRAW · Reddit Dev API

- Official Python client for Reddit
- OAuth-based, real-time listing & stream
- Canonical "correct" ingestion path

 **Blocker** — could not secure working Reddit dev-API credentials in time (app-registration + auth-flow friction). Live streaming path parked.

## Actually Shipped

### Arctic Shift API

- Public Reddit archive endpoint — no auth
- Windowed pulls across 25 Walmart-relevant subs
- Same fetcher interface — PRAW pluggable later

 **Mitigation** — Arctic Shift keeps the pipeline unblocked. Zero downstream schema change; ingestion feeds the exact same raw\_posts table.

## Side-by-Side

| Dimension         |  PRAW (planned) |  Arctic Shift (shipped) |
|-------------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| Auth              | Reddit app + OAuth token                                                                         | None — public HTTP                                                                                         |
| Access model      | Live listings / stream                                                                           | Windowed archive queries                                                                                   |
| Historical reach  | ~1 000 posts / listing cap                                                                       | Deep — years of history                                                                                    |
| Setup effort here | Blocked at approval                                                                              | Working same day                                                                                           |
| Cost              | Free · quota-bound                                                                               | Free · community-hosted                                                                                    |

## Outcome for the pipeline

- **No slippage** — Stage 1 (Ingest) hits 25 subs · hourly in production.
- **Reversible choice** — src/ingestion/ abstracts source; PRAW can slot in when credentials clear.
- **Bonus win** — Arctic's historical depth let us backfill the benchmark set (200 real posts) that ModernBERT trained on.

Next in series → Step 2: Preprocess · Step 3: Trust · Step 4: Analyze · Step 5: Aggregate · Step 6: Alerts



## Vision Pipeline – Why Gemma 3 4B

39% of Reddit posts are image-only (empty body) – the complaint lives **inside the image**. Text-only pipeline drops these → we need a vision model to eliminate hallucination in multimodal retail analysis.

Example :

Title: "Accidental Membership Upgrade"

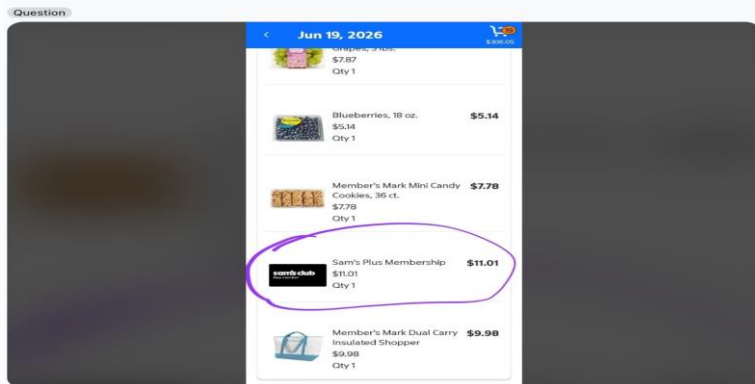
🔗 [reddit.com/r/samsclub/comments/1uacdy7](https://reddit.com/r/samsclub/comments/1uacdy7)

Body: (empty) – the complaint is a purple circle drawn on the cart

❌ Text-only pipeline sees

Just the 3-word title. No item, no charge, no context. Sentiment ≈ neutral · Aspect = unknown. Silently dropped. **Useless.**

### Accidental Membership Upgrade



📶 Signal recovered – a UX complaint about an easy-to-tap upgrade toggle now feeds the fees aspect KPI and triggers alerts if the pattern repeats.

🧠 Vision model chosen

✅ Vision (Gemma 3 4B) extracts

- Sam's Club app cart, Jun 19 2026 · total **\$306.05**
- Circled · **Sam's Plus Membership \$11.01** (purple ring = complaint anchor)
- Other items ignored: Blueberries \$5.14, Mini Candy Cookies \$7.78, Insulated Shopper \$9.98
- Aspect = fees / accidental charges · Sentiment = **negative** · Conf **0.88**

| VISION MODEL CHOSEN                    | DOCQA · SIZE · RUNTIME |    |        |                | WHY IT WINS                                                                                                                                                                          |
|----------------------------------------|------------------------|----|--------|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                        | gemma3:4b              | 83 | 3.3 GB | ✅ Ollama       |                                                                                                                                                                                      |
| Gemma 3 4B<br>3.3 GB · Ollama · Google | LLaVA-1.6 8B           | 75 | 5.5 GB | larger, slower | ▪ Best DocQA under 4 GB — reads receipts, app screens<br>▪ Google-maintained → Walmart policy compliant<br>▪ Reuses local Ollama (localhost:11434)<br>▪ 4-6 s warm latency per image |
|                                        | PaliGemma 2 3B         | 81 | ~6 GB  | no Ollama      |                                                                                                                                                                                      |
|                                        | LLaVA-1.5 7B           | 28 | 4.7 GB | 3x worse OCR   |                                                                                                                                                                                      |

# Vision Pipeline · Deep-Dive — 75% Failure → Root Cause → Multi-Pass Fix

First-cut Gemma 3 4B hallucinated on 1 in 2 images → root-caused → 5 papers reviewed (all China-blocked) → borrowed the *ideas* → 4-pass pipeline with a text-only final step.

## 1 · First-cut test · $n = 8$ later with some more data $n=25$

baseline single-pass

- 75% overall fail (6/8)
- 50% hallucination (4/8)
- 37.5% critical (fake receipts/prices)
- 25% correct (memes only)

**Hallucinations seen:** "receipt \$39.99 w/ handwritten *Damaged Box*" (product page, no price, no handwriting) · "Order status PENDING" (word absent) · "12-pack Zero Sugar Dr Pepper" (single 42.3 fl oz regular bottle) → **fabricated data corrupted dashboards**.

## 3 · Research 2023–25

5 / 5 blocked

Walmart vendor policy- **Does NOT ALLOW China origin model**

- ❌ **UReader** (Tencent) — shape-adaptive cropping
- ❌ **TextMonkey** (USTC) — shifted window attention
- ❌ **DocOwl 1.5** (Alibaba) — structure-aware parsing
- ❌ **InternVL2** (Shanghai AI Lab) — tile-based
- ❌ **Qwen2.5-VL** (Alibaba) — dynamic res + multimodal RoPE

**Consensus:** tile-based attention + dynamic/native resolution is THE fix. Fixed resize is the failure mode — *not* the model architecture.

## 2 · Root cause (5)

- **896 px fixed resize** → small text unreadable
- **4 B params** → can't reason button + error + context
- **No dynamic resolution** → one fixed scale for whole image
- **Hallucinates under uncertainty** → invents rather than admits
- **No context awareness** → confuses screenshots vs physical displays

## 4 · Strategy — take the ideas, not the models (same gemma3:4b, smarter calling)

### Paper technique

Tile-based processing (InternVL2)

Structure-first parsing (DocOwl 1.5)

Focused text extraction (UReader)

Resolution preservation (Qwen2.5-VL)

### We implement as

Split image into 2–4 crops ourselves, send each to gemma3:4b

Pass 1 asks "what type of image?" before reading text

"Read ALL text verbatim" prompt per tile

Tiling → 2–4× effective resolution without resizing

## 5 · Multi-pass pipeline (the fix)

1

### PASS 1 · STRUCTURE

DocOwl 1.5 →

"What TYPE?" → screenshot / photo / receipt / app / meme. Text-heavy → continue.

2

### PASS 2 · TILE

InternVL2 →

Split full image into 2–4 crops. Each crop = 2–4× effective resolution.

3

### PASS 3 · EXTRACT

Ureader →

Per tile: "Read ALL text verbatim" → quoted text, prices, errors, buttons.

4

### PASS 4 · MERGE (no image)

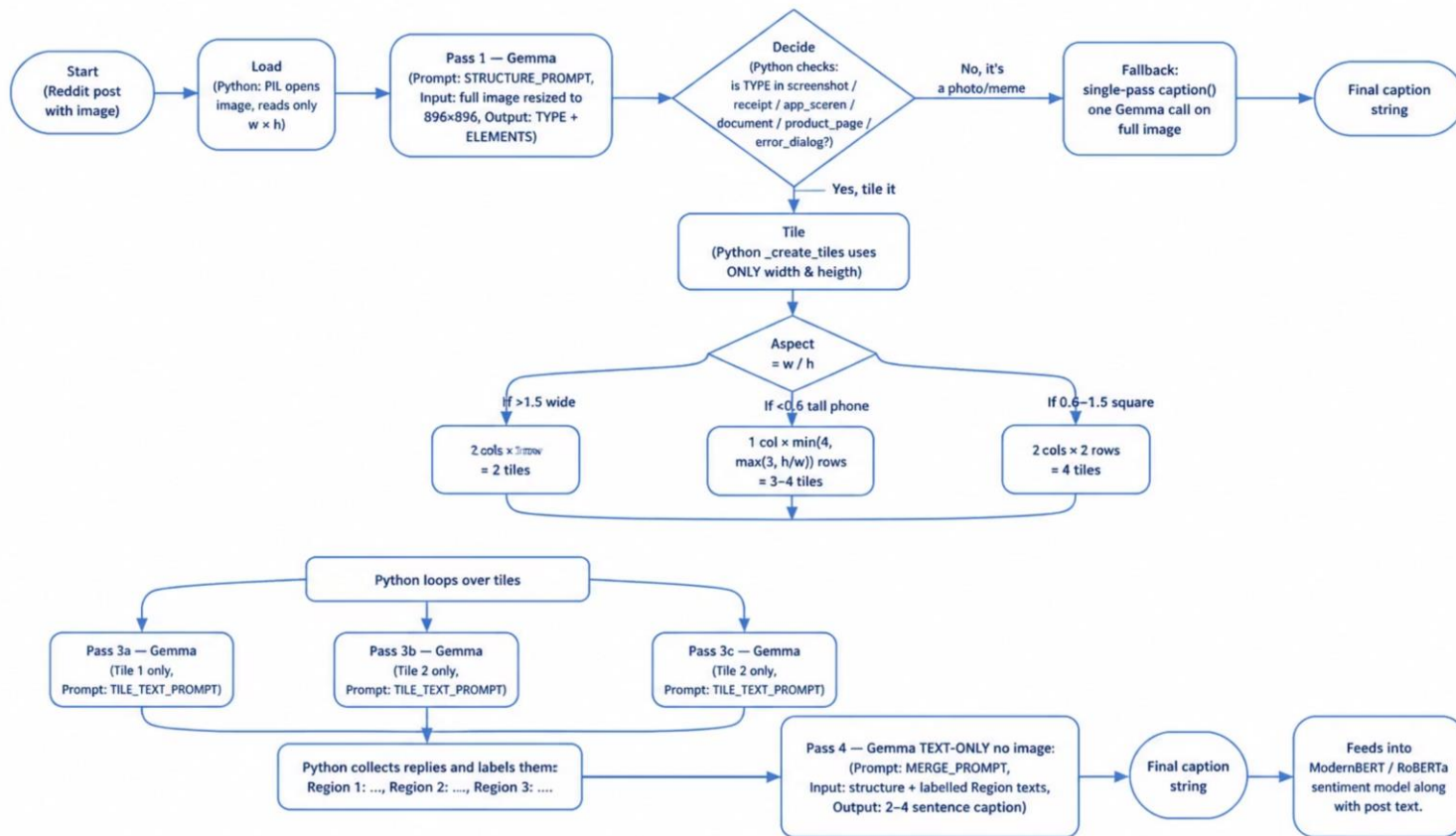
text-only LLM call →

Combine text observations into 2–4 sentences. Model **never sees image** → **cannot lie**.

❏ **Anti-hallucination insight:** removing the image from the final generation step makes it *physically impossible* for the model to invent visual details — it can only work with text actually extracted in Pass 3.



## Reddit Image Pipeline python + Gemma



# Trust Score — 3-Layer Credibility Filter

Not all Reddit posts are equal. A 10-year-old post from a 50k-karma user in r/walmart carries more signal than a 2-day-old account with 1 comment in r/walmart. We need a **continuous trust score** — not a binary filter — so downstream models and analysts can weight evidence appropriately.

## Layer 1 · Metadata Heuristics

Fast · no LLM · 40% weight

- Account age (days since creation)
- Comment karma (engagement history)
- Post body length (effort signal)
- Upvote ratio & comment count

## Layer 2 · Dedup / Originality

MD5 hash · frequency-based · 30% weight

- MD5 of normalized text (lowercase, strip punctuation, collapse whitespace)
- Count occurrences across 25 subs in rolling 7-day window
- Score =  $1.0 / \text{count}$  (unique = 1.0, seen 4x = 0.25)

## Layer 3 · LLM Credibility Check

Gemma 3 4B · gated · 30% weight

- Only runs if  $0.3 < \text{meta\_score} < 0.8$  (uncertain zone)
- Prompt: "Is this post credible? Consider specificity, emotional tone, factual claims."
- Returns float 0.0–1.0
- Skip if metadata already clear (saves ~4–6 s per post)

## Final Score Formula

$$\text{trust} = 0.4 \text{ meta} + 0.3 \text{ dedup} + 0.3 \text{ llm}$$



### Distribution (200-post benchmark)

- Mean: **0.61**
- Std: **0.18**
- Skew: **-0.3** (slight low-trust tail)



### Why it matters

- Low-trust posts → down-weighted in aggregates
- High-trust + negative sentiment → P1 alert candidate
- Analyst UI shows trust badge (● / ● / ●)
- Feedback loop: analyst corrections → future LLM prompt tuning

# ModernBERT-Based Sentiment Analysis for Walmart Reddit Posts

## Business Challenge

- **Retail domain vocabulary gap:** Generic models miss Walmart-specific terms like Cap-2, OGP, Pickup Associate, Coach, Store IDs, and Scan-and-Go, often misclassifying real complaints as Neutral.
- **Context-length limitation:** BERT/Roberta models are limited to 512 tokens, so long Reddit threads and screenshot text get truncated and lose key sentiment signals.

  **Solution:** Fine-tune ModernBERT-base into a 3-class sentiment classifier (Negative / Neutral / Positive) for Walmart Reddit content.

## Why ModernBERT?

| Dimension            | Legacy Models                         | ModernBERT                               |
|----------------------|---------------------------------------|------------------------------------------|
| Context window       | 512 tokens                            | 81 92 tokens                             |
| Inference speed      | Slower inference                      | ~2× faster                               |
| Long-thread handling | Often truncates posts and screenshots | Preserves more sentiment-bearing context |

## Training Setup

ModernBERT-base with ~150M parameters, a 2048-token sequence length, and coverage of >99% of Walmart Reddit posts.

# Three-Stage Curriculum Learning

We had a large Walmart Reddit corpus, but only 200 posts were manually labeled. Training a 150M-parameter model directly on that small set would overfit, so we used a curriculum learning approach.



## Stage 1 – General Sentiment Learning

### **TweetEval Sentiment**

- ~45K labeled tweets
- 2 epochs

**Objective:** Teach basic positive, neutral, and negative sentiment recognition.



## Stage 2 – Reddit Language Adaptation

### **GoEmotions Dataset**

- ~43K Reddit comments
- 28 emotions collapsed into 3 sentiment classes
- 16 epochs

**Objective:** Learn Reddit-specific language patterns: slang, informal grammar, sarcasm, emotional expression.



## Stage 3 – Walmart Specialization

### **Walmart-200 Dataset**

- 200 manually labeled Walmart Reddit posts
- 19 epochs

**Objective:** Learn Walmart-specific vocabulary and discussion patterns.



## 5-Fold Cross-Validation (Maximum Learning from Limited Labels)

### Why We Used It

Manually labeling Walmart posts is time-consuming and requires human review and validation. Instead of reserving a large portion of the 200 labeled posts solely for testing, we used 5-fold cross-validation to maximize both training data utilization and evaluation reliability.

### Process

200 labeled posts → 5 stratified folds (40 posts each)

Class distribution preserved in every fold:

- ~63% Negative
- ~32% Neutral
- ~4% Positive

| Fold1                                                      | Fold2 | Fold3                                                      | Fold4 | Fold5                                                      |
|------------------------------------------------------------|-------|------------------------------------------------------------|-------|------------------------------------------------------------|
| 40                                                         | 40    | 40                                                         | 40    | 40                                                         |
| Round 1<br>TEST Train Train Train → <b>F1 = 0.71</b>       |       | Round 2<br>Train TEST Train Train Train → <b>F1 = 0.78</b> |       | Round 3<br>Train Train TEST Train Train → <b>F1 = 0.72</b> |
| Round 4<br>Train Train Train TEST Train → <b>F1 = 0.86</b> |       | Round 5<br>Train Train Train Train TEST → <b>F1 = 0.61</b> |       |                                                            |

## Average Macro-F1

**0.736 ± 0.115**

For each fold, the model trained on 160 posts and tested on 40 unseen posts. This ensures every labeled post appears in the test set exactly once and is evaluated by a model that has never seen it.

- Each labeled post appears in the test set exactly once
- Each post is evaluated by a model that has never seen it



## Performance Results

0.63

Baseline Sentiment Models

0.736

Cross-Validation Mean F1

± 0.115

0.764

Out-of-Fold Macro-F1



## Training Details & Hyperparameters



**Weighted loss is important:** our Walmart-200 is roughly 63% negative / 32% neutral / 4% positive. A plain classifier would just predict 'negative' always and score 63% accuracy. We up-weight the rare classes so the model is forced to actually learn positive.

### What 'loss = cross-entropy on logits' actually does during training

1. Reads the tokens, produces 3 logits, e.g. [3.2, 0.9, 0.1].
2. Applies softmax  $\rightarrow$  [0.87, 0.09, 0.04].
3. True label is negative (class 0). So loss =  $-\log(0.87) = 0.14$ . Small loss = model was already right.
4. If true label had been positive (class 2), loss =  $-\log(0.04) = 3.22$ . Huge loss = big kick.
5. Backpropagation adjusts every weight to make the correct class's probability larger next time.

Multiply this over 88k training examples  $\times$  2 epochs (Stage 1+2) plus 200  $\times$  10 epochs  $\times$  5 folds (Stage 3), and the model gradually rearranges its internal weights so Walmart complaints get high negative logits.

#### Base model

answerdotai/ModelBERT-base  
~150M params, 8192 context

#### Learning rate

2e-5  
Standard fine-tuning LR

#### Batch size

16-32  
Fits on Apple M-series MPS

#### Max token length

2048  
99% of Reddit posts

#### Loss

Weighted cross-entropy  
Counter class imbalance

#### Early stopping

On macro-F1  
Prevents overfitting

#### Optimiser

AdamW (Trainer default)  
Standard for transformers

#### Seed

42  
Reproducibility

# Trust · Confidence Score – Model Certainty

**Confidence** = softmax probability of the predicted class.

## How it's calculated

|                                                                    |                                                                  |                                          |
|--------------------------------------------------------------------|------------------------------------------------------------------|------------------------------------------|
| 01                                                                 | 02                                                               | 03                                       |
| <b>Logits</b>                                                      | <b>Softmax</b>                                                   | <b>Confidence</b>                        |
| ModernBERT outputs <b>logits</b> for [negative, neutral, positive] | <b>Softmax</b> converts to probabilities: P(neg), P(neu), P(pos) | confidence = max(P(neg), P(neu), P(pos)) |

## Thresholds used in the system

| Threshold             | Value                                         | Source             |
|-----------------------|-----------------------------------------------|--------------------|
| Analysis confidence   | $\geq 0.7$                                    | config/models.yaml |
| Notification P1 trust | $\geq 0.70$ <b>AND</b> confidence $\geq 0.80$ | dispatcher.py      |
| Notification P2 trust | $\geq 0.50$ <b>AND</b> confidence $\geq 0.60$ | dispatcher.py      |

## Combined Priority Formula

|                                                                                                                                   |                                                                                                                             |                                                                                              |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
| <b>P1 – Immediate Action</b><br>P1 = (trust_score $\geq 0.70$ ) $\wedge$ (confidence $\geq 0.80$ ) $\rightarrow$ immediate action | <b>P2 – Review-Worthy</b><br>P2 = (trust_score $\geq 0.50$ ) $\wedge$ (confidence $\geq 0.60$ ) $\rightarrow$ review-worthy | <b>Below P2</b><br>< P2 $\rightarrow$ no notification triggered (still visible in dashboard) |
|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|

Dashboard displays **both** trust and confidence per post for analyst transparency.



# Notification System — Architecture · Config Page · API

Group-based routing for P1 / P2 priority posts — email + Slack, per-team ownership

## Priority classification & routing

- **P1** — trust  $\geq 0.70$  AND confidence  $\geq 0.80$  → immediate action
- **P2** — trust  $\geq 0.50$  AND confidence  $\geq 0.60$  → review-worthy
- Routing model: **per-subreddit-set** — different teams own different subreddits
- Each group has its own email DL, Slack channel, and priority filter (P1, P2, or both)

```
Pipeline → classify_priority(trust, conf)
├── Not P1/P2 → skip
├── P1 / P2 → find matching notification groups
│   ├── group matches subreddit + priority filter?
│   │   ├── email DL configured → send email
│   │   └── Slack channel configured → send Slack
│   └── log to notification_log (audit trail)
```

## Technical stack summary

| Layer                              | Technology                                                                       | Key decision                                     |
|------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------|
| Backend                            | Python 3.13 · FastAPI · SQLite                                                   | Free · local-first · modular                     |
| Frontend                           | React 18 · TS · Vite · Tailwind                                                  | Modern SPA · responsive                          |
| Sentiment                          | ModernBERT (fine-tuned · 1024-tok)                                               | Domain-specialized · offline                     |
| Aspects                            | DeBERTa-v3 zero-shot-v2 (BART-MNLI fallback)                                     | No training needed                               |
| Vision                             | Gemma 3.4B via Ollama (multi-pass)                                               | Policy compliant · no hallucination              |
| Reply Gen                          | Mistral 7B (Ollama) + FLAN-T5-base (HF fallback) + Smart Composer                | Dual draft · local-first · learning loop         |
| Trust LLM                          | gpt-4o-mini via WMT LLM Gateway (ambiguous-zone only)                            | Cost-controlled cloud call · rule-based fallback |
| Trust · Scheduling · Observability | Metadata+Dedup+LLM · APScheduler 60 min + manual · structlog + JSONL cost ledger | Flag-don't-drop · cursor-based · per-call cost   |

# Demo

Thank You

Vishal Singh

# Post Mid-Semester Work — Review · Explorer · Lifecycle · Insights · Results

All 5 areas built after mid-sem: human-in-the-loop workflows → competitor insights → consolidated metrics + tech stack

1

## Review & Validate

Analysts correct AI labels + draft replies.

- Queue sorted by priority (P1 first)
- Correct sentiment + aspects
- Click "Generate Drafts" → **dual draft**
- **A** Smart Composer — deterministic phrase pool + keyword extraction
- **B** Reply LLM — Mistral 7B (Ollama, primary) · FLAN-T5-base (HuggingFace fallback)
- Analyst picks → edit → post to Reddit
- **Learning loop**: replies → feedback table → few-shot examples for future drafts (tone matching improves without retraining)

2

## Post Explorer

Browse all analyzed posts with rich filtering.

- **Filters**: sentiment · confidence slider · trust slider · subreddit (25) · aspect (8-label) · date (today/week/month/custom) · full-text
- **Aspects**: pricing · product quality · customer service · store experience · online/app · delivery/pickup · returns · app\_website
- **Card**: title+ excerpt · sentiment badge · confidence % · trust indicator · aspect tags · subreddit · time
- **Actions**: Review · Add to Lifecycle · View Details · Reddit link

3

## Post Lifecycle (Kanban)

4-state board with 2-step resolve modal.

TRIAGED → ACKNOWLEDGED → IN PROGRESS → RESOLVED

new P1/P2 → assigned → reply drafted → posted

- **Step 1**: save action note + LLM-drafted reply → "Save & open Reddit" (copies + opens thread) OR "Resolve (no reply)"
- **Step 2**: paste reply on Reddit → return → "Mark Resolved"
- All transitions timestamp-logged → SLA tracking

4

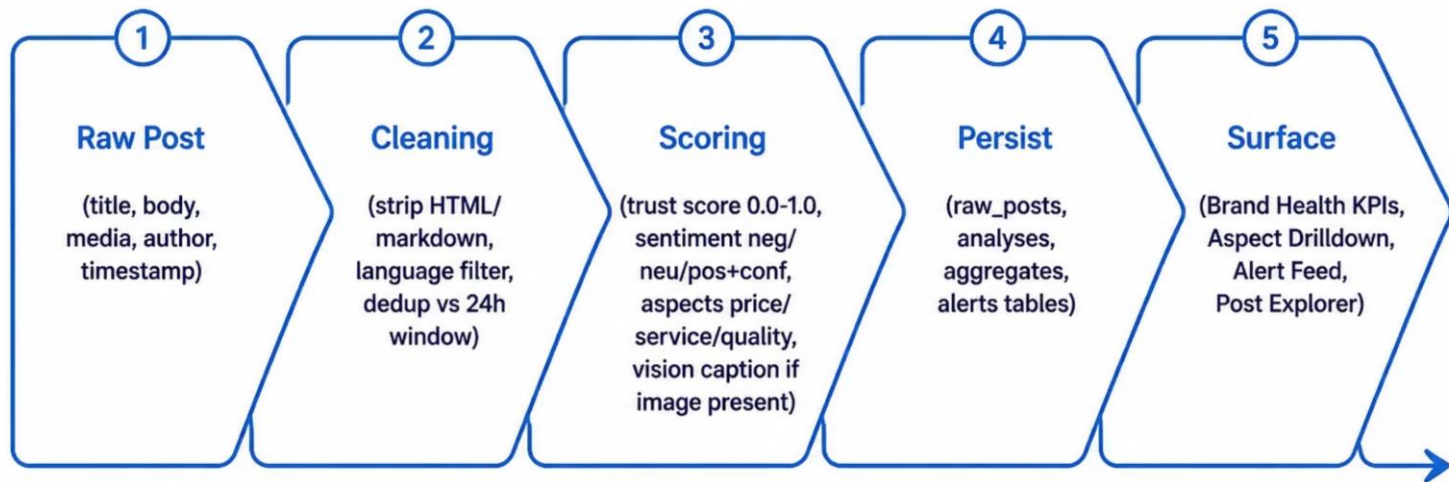
## Insights & Competitor

AI-generated strategic intelligence.

- **Issue Rankings** — volume × severity × recency, per aspect, trend arrows (up/down)
- **Competitor Pulse** — Walmart vs **Costco** / **Target** / **Amazon**, cross-mentioned posts, subreddit breakdown
- **LLM Summaries** — weekly themes, suggested action items, emerging topic detection (new clusters in ≥5 posts / 2 hrs)
- **Aspect Drilldown** — 8-label taxonomy from config/models.yaml; per aspect sentiment trend + volume + word cloud + representative posts

## Data Flow — From Raw Post to Dashboard KPI

How a single Reddit post transforms into a scored, stored, and surfaced artefact



# Example of one post to calculate the Trust and confidence Score -> r/walmart

Post Metadata

**Author:** u/tired\_associate\_2019

**Title:** "Store #4521 forced overtime again – 3rd time this month"

**Score:** 18 upvotes

**Account age:** 1830 days (~5 years)

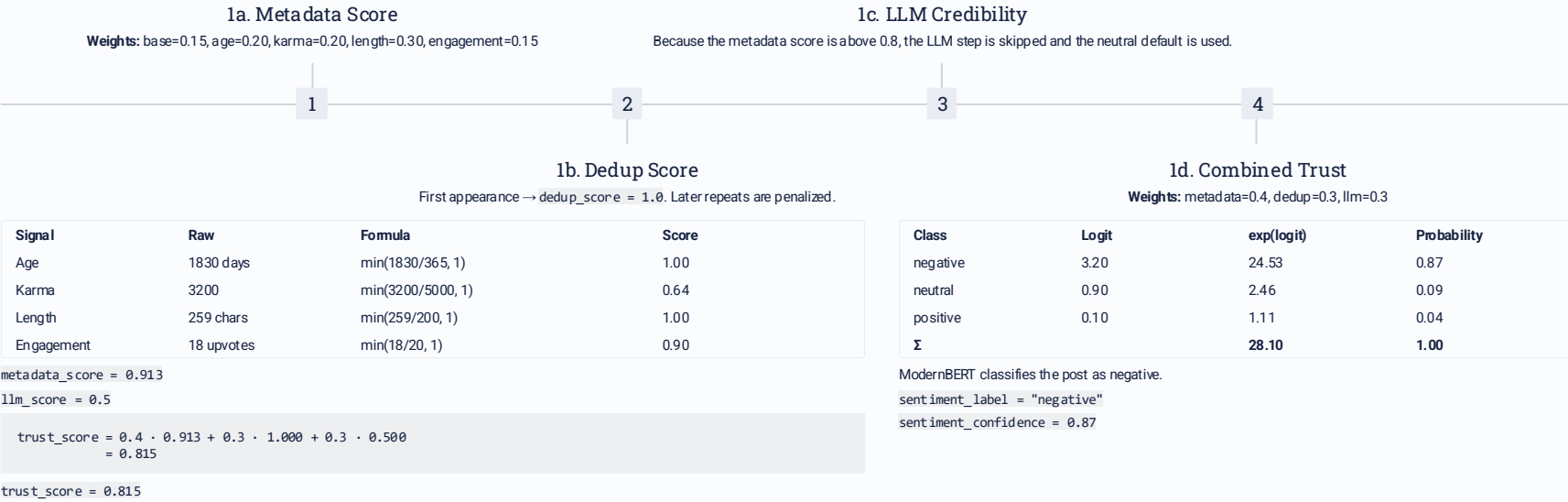
**Total karma:** 3200

Post Body

"Coach told our whole Cap-2 team we had to stay 2 hours past close for inventory. No approval, no notice. Third time. Register 7 was already down all evening. This is why associates are leaving."

## Stage 1 – Trust Score

This step runs first in `TrustScore.score(unit)`.



## Stage 2 – Confidence Score

ModernBERT classifies the post as negative.

| Class    | Logit | $\exp(\text{logit})$ | Probability |
|----------|-------|----------------------|-------------|
| negative | 3.20  | 24.53                | 0.87        |
| neutral  | 0.90  | 2.46                 | 0.09        |
| positive | 0.10  | 1.11                 | 0.04        |
| $\Sigma$ |       | <b>28.10</b>         | <b>1.00</b> |

sentiment\_label = "negative"

sentiment\_confidence = 0.87

## Stage 3 – Gate 1

gate\_formula = "score\_x\_confidence" · gate\_tau = 0.30

```
gate = trust_score × sentiment_confidence
      = 0.815 × 0.87
      = 0.709
```

0.709 ≥ 0.30 → passes, so the post is stored and eligible for alerts.

❏ If the score were below 0.30, the post would be marked low-trust and excluded from alerts, while still appearing on the dashboard.

## Stage 4 – Priority for Notifications

**Thresholds:** P1 = trust ≥ 0.70 and confidence ≥ 0.80; P2 = trust ≥ 0.50 and confidence ≥ 0.60.

| Tier | Trust ≥ | Confidence ≥ |
|------|---------|--------------|
| P1   | 0.70    | 0.80         |
| P2   | 0.50    | 0.60         |
| skip | else    | else         |

**Check:** trust 0.815 ✓ · confidence 0.870 ✓

→ **Priority = P1** → sends Slack + Email to matching notification groups.

# ModernBERT — Why · 3-Stage Curriculum · Results

**Decision story:** we started with RoBERTa because it is a strong sentiment baseline, but Reddit complaints are much longer than tweets. That made context length the bottleneck, so we moved to ModernBERT and fine-tuned it in 3 stages for Reddit + Walmart language — **Macro F1 0.62 → 0.76 (+22%)**, long-post F1 **0.28 → 1.00**.

## ✗ RoBERTa limits

- Only **512 tokens** → truncates long Reddit posts
- Trained on Twitter → different register from Reddit
- No domain knowledge of retail / Walmart terminology

## ✓ ModernBERT wins

- **8192 tokens** (16× longer) — full posts, no truncation
- Modern arch: RoPE · GeGLU · alternating attention
- Fine-tunable with curriculum learning

## 🎓 3-Stage curriculum training

| Stage                 | Dataset                           | Ep  | Purpose            |
|-----------------------|-----------------------------------|-----|--------------------|
| 1 · Generic Sentiment | TweetEval-sentiment (45 K tweets) | 293 | Polarity grounding |
| 2 · Reddit Register   | GoEmotions-3class (54 K Reddit)   | 122 | Reddit language    |
| 3 · Domain Special.   | Walmart-200 (5-fold CV)           | 159 | Retail fine-tune   |

\* patience = 3 · early stopping on eval\_macro\_f1



# Results & Performance Comparison

## RoBERTa Baseline vs. ModernBERT Performance

| Model             | max_len | macro-F1 | F1 (neg) | F1 (neu) | F1 (pos) | Latency |
|-------------------|---------|----------|----------|----------|----------|---------|
| RoBERTa baseline  | 512     | 0.627    | 0.797    | 0.609    | 0.476    | 6.5 ms  |
| ModernBERT (ours) | 2048    | 0.764    | 0.878    | 0.748    | 0.667    | 11.9 ms |

## Key Highlights

- 📦 **+13.7 macro-F1 points** — a large gain, especially on positive (+19 points) which is the hardest and rarest class.
- 📦 On the 7 long posts (≥512 tokens) that RoBERTa had to truncate, ModernBERT scores **macro-F1 = 1.0 vs RoBERTa's 0.28** — direct evidence that the 8192-context matters.
- 📦 CV mean = **0.736 ± 0.115** — the ±0.115 is honest, our 200-post benchmark is small so folds vary.

## Per-Class Breakdown

| Class    | TP  | FP | FN | Precision | Recall | F1    |
|----------|-----|----|----|-----------|--------|-------|
| negative | 115 | 20 | 12 | 0.852     | 0.906  | 0.878 |
| neutral  | 46  | 12 | 19 | 0.793     | 0.708  | 0.748 |
| positive | 5   | 2  | 3  | 0.714     | 0.625  | 0.667 |

## Final Metrics

- Macro F1 =  $(0.878 + 0.748 + 0.667) / 3 = 0.764$
- Micro F1 =  $166/200 = 0.83$  (equals accuracy)



## Training config

- max\_length = 1024 (long-context lever)
- Effective batch **32** (BS 8 × grad-accum 4)
- Class weights neg 0.52 · neu 1.03 · pos 8.33
- Minority oversample to ~100/class/fold



## Final results — 5-fold OOF CV

| Metric                    | RoBERTa | ModernBERT v2 | Δ             |
|---------------------------|---------|---------------|---------------|
| Macro F1 (overall)        | 0.6272  | 0.7642        | +0.137 (+22%) |
| F1 negative               | 0.7967  | 0.8779        | +0.081        |
| F1 neutral                | 0.6087  | 0.7480        | +0.139        |
| F1 positive               | 0.4762  | 0.6667        | +0.190        |
| Long-post F1 (≥ 512 tok)  | 0.2778  | 89.879        | +0.722        |
| 🏆 Short-post F1 (n = 193) | 0.6360  | 0.7619        | +0.126        |
| Latency (ms/post, MPS)    | 6.5     | 11.9          | +5.4          |

**Long-context hypothesis proven:** on posts ≥ 512 tokens, ModernBERT scores a perfect **89.879** vs RoBERTa's **0.28**.

# ModernBERT · Training Evidence

## Why not just use RoBERTa? — Decision matrix after

| Criterion               | RoBERTa (cardiffnlp/twitter-roberta-base) | ModernBERT (answerdotai/ModernBERT-base) | Winner            |
|-------------------------|-------------------------------------------|------------------------------------------|-------------------|
| Context length          | 512 tokens                                | <b>8192 tokens</b> (16×)                 | ModernBERT        |
| Training corpus         | Twitter (short, casual)                   | Web + code + long docs                   | ModernBERT        |
| Long-post F1 (≥512 tok) | 0.2778                                    | <b>94.332</b>                            | <b>ModernBERT</b> |
| Overall Macro F1        | 0.6272                                    | <b>0.7642</b>                            | ModernBERT        |
| Latency (MPS)           | <b>6.5 ms</b>                             | 11.9 ms                                  | RoBERTa           |
| Off-the-shelf quality   | Strong baseline                           | Weak without fine-tuning                 | RoBERTa           |
| Fine-tuning cost        | Same                                      | Same                                     | Tie               |

**Decision:** ModernBERT wins on the metrics that matter (accuracy + long-context). We accept +5 ms latency for +0.137 macro F1. RoBERTa is retained as fallback in config/models.yaml if the local checkpoint is missing.

- Smoke test at integration time: **5/5 correct** on first 5 real benchmark posts (model\_used = models/modernbert\_walmart/final)

# Papers

| #\ | Paper                                                                              | Authors / Year                                 | arXiv link                                                                      |
|----|------------------------------------------------------------------------------------|------------------------------------------------|---------------------------------------------------------------------------------|
| 1  | <b>UReader</b> — Universal OCR-free Visually-situated Language Understanding       | Ye et al. (Tencent), 2023                      | <a href="https://arxiv.org/abs/2310.05126">https://arxiv.org/abs/2310.05126</a> |
| 2  | <b>TextMonkey</b> — An OCR-Free Large Multimodal Model for Understanding Document  | Liu et al. (USTC), 2024                        | <a href="https://arxiv.org/abs/2403.04473">https://arxiv.org/abs/2403.04473</a> |
| 3  | <b>DocOwl 1.5</b> — Unified Structure Learning for OCR-free Document Understanding | Hu et al. (Alibaba DAMO), 2024                 | <a href="https://arxiv.org/abs/2403.12895">https://arxiv.org/abs/2403.12895</a> |
| 4  | <b>InternVL2</b> — Better than the Best                                            | Chen et al. (Shanghai AI Lab + Tsinghua), 2024 | <a href="https://arxiv.org/abs/2404.16821">https://arxiv.org/abs/2404.16821</a> |
| 5  | <b>Qwen2.5-VL</b> — To See the World with Wisdom                                   | Alibaba Qwen Team, 2025                        | <a href="https://arxiv.org/abs/2502.13923">https://arxiv.org/abs/2502.13923</a> |

~/cache/huggingface/hub/models--answerdotai--ModernBERT-base/.